CALCULUS I - AY2024-25

Chapter 4: Integrals and Applications

Outline

1. Indefinite integral
2. Definition of the definite integral
3. Properties of the definite integral
4. Fundamental Theorem of Calculus
5. Methods of Substitution and Integration by Parts
6. Volumes
7. Arc Length and Surface Area

Summary

We have studied in Chapter 3 antiderivative of a function and the indefinite integral as its collection of all antiderivatives.

In this section we develop some fundamental results about antiderivatives and indefinite integrals. The main topics are:

- Properties of indefinite integrals.
- Two important integration methods.

From the Antiderivative linearity rules, we also obtain the following.

Theorem (Properties of the Indefinite Integral)

Suppose $F(x)$ and $G(x)$ are antiderivatives of $f(x)$ and $g(x)$ respectively, and k is a constant. Then

- *A constant factor can be moved through an integral sign, i.e.*

$$\int kf(x)\,dx = k\int f(x)\,dx = kF(x) + C$$

- *An integral of a sum/difference is the sum/difference of integrals, i.e.*

$$\int \left[f(x) \pm g(x)\right]\,dx = \int f(x)\,dx \pm \int g(x)\,dx = F(x) \pm G(x) + C.$$

Here C is an arbitrary constant.

• **Integration by Substitution**

The chain rule gives the method integration by substitution

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

Here we substitute $u = g(x)$ and $du = g'(x)\,dx$ and so this method is also called the method of u-substitution.

Example

Find $\int x^3 \cos(x^4 + 2)\, dx$.

Solution. We put $u = x^4 + 2$ which gives $du = 4x^3\, dx$. Then since $x^3\, dx = \frac{1}{4}\, du$, we have

$$\begin{aligned} \int x^3 \cos(x^4 + 2)\, dx &= \int \cos u \cdot \frac{1}{4}\, du = \frac{1}{4} \int \cos u\, du \\ &= \frac{1}{4} \sin u + C = \frac{1}{4} \sin(x^4 + 2) + C, \end{aligned}$$

where C is an arbitrary constant.

• Integration by Parts

The derivative rule for product gives the method of integration by parts

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\,dx;$$

which is easier to remember in the following notation

$$\int u\,dv = uv - \int v\,du,$$

with $u = f(x)$ and $v = g(x)$.

Example

Find $\int x \sin x \, dx$.

Solution. We put

$$\begin{cases} u = x \\ dv = \sin x \, dx \end{cases} \Longrightarrow \begin{cases} du = dx \\ v = -\cos x. \end{cases}$$

Then we have

$$\int x \sin x \, dx = -x \cos x + \int \cos x \, dx = -x \cos x + \sin x + C,$$

where C is an arbitrary constant.

From the Table of antiderivative formulas we obtain the following "Table of Integration Formulas".

1. $\int k\,dx = kx + C$ (any number k)
2. $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$ $(n \neq -1)$
3. $\int \frac{dx}{x} = \ln|x| + C$
4. $\int e^x\,dx = e^x + C$
5. $\int a^x\,dx = \frac{a^x}{\ln a} + C$ $(a > 0, a \neq 1)$
6. $\int \sin x\,dx = -\cos x + C$
7. $\int \cos x\,dx = \sin x + C$
8. $\int \sec^2 x\,dx = \tan x + C$
9. $\int \csc^2 x\,dx = -\cot x + C$
10. $\int \sec x \tan x\,dx = \sec x + C$
11. $\int \csc x \cot x\,dx = -\csc x + C$
12. $\int \tan x\,dx = \ln|\sec x| + C$
13. $\int \cot x\,dx = \ln|\sin x| + C$
14. $\int \sec x\,dx = \ln|\sec x + \tan x| + C$
15. $\int \csc x\,dx = -\ln|\csc x + \cot x| + C$
16. $\int \sinh x\,dx = \cosh x + C$
17. $\int \cosh x\,dx = \sinh x + C$
18. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$
19. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$
20. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{x}{a}\right| + C$
21. $\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C$ $(a > 0)$
22. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C$ $(x > a > 0)$

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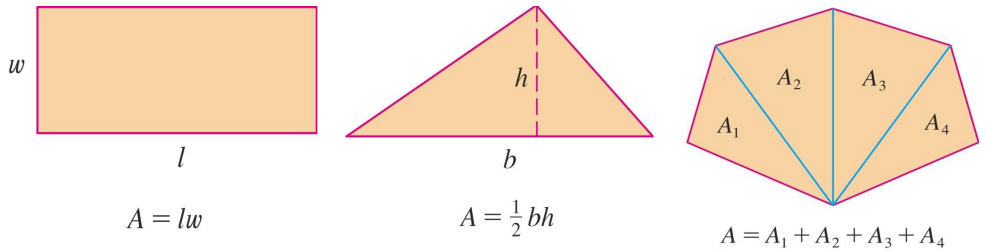
Summary

In this section we study the following.

- How to define the value of the area under a curve.
- What a Riemann sum is.
- What the definite integral of a function is.

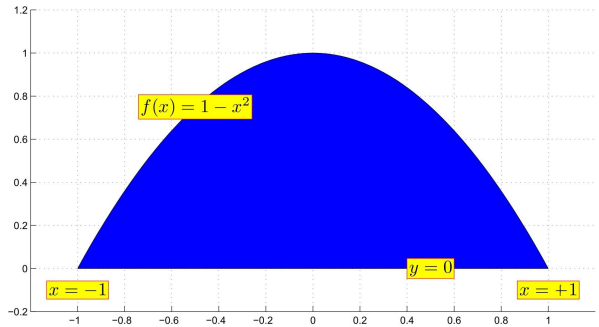
Given a figure bounded by some curve, how can we find its area?

- If it is a “regular” figure, then the answer is easy



- However, what happens if the region is bounded by a curve $y = f(x)$?

For example, a region is bounded by the curve $y = f(x) = 1 - x^2$, the x-axis, and the lines $x = \pm 1$.



- “Integration", or “integral calculus" is the formal mathematical method used to define the area of such a region and to develop methods to help in the calculations.

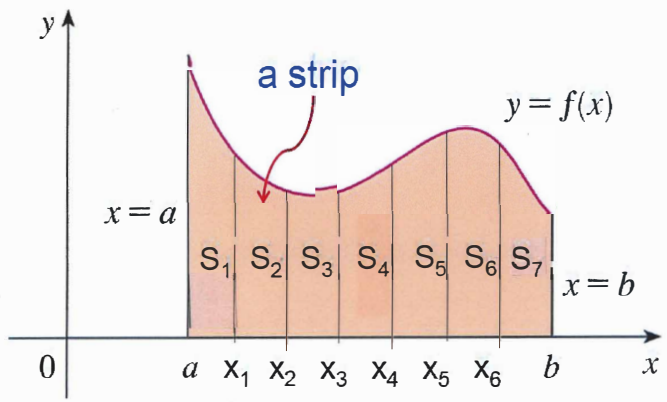
The idea is to “approximate" the unknown areas using regular figures (such as rectangles) and then apply the limit.

We approximate this area with a collection of strips as shown below.

We divide the given interval $[a, b]$ into n equal subintervals. Let

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$$

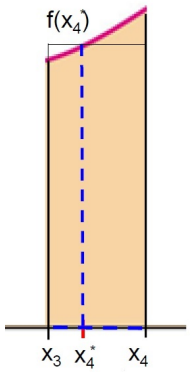
In the figure we have 7 strips, so $n = 7$. The seven intervals are $[a, x_1], [x_1, x_2], \cdots, [x_6, b]$.



The trick is to approximate this area with a rectangular strip.

- Let's consider the area over say interval $[x_3, x_4]$. The width of the rectangle is the same as the width of the strip $x_4 - x_3$.
- However we need to specify the height of the approximating rectangle. For this, we choose a sampling point in the subinterval, say $x_4^* \in [x_3, x_4]$ and look at the rectangle with height $f(x_4^*)$.

This is illustrated in the figure below.



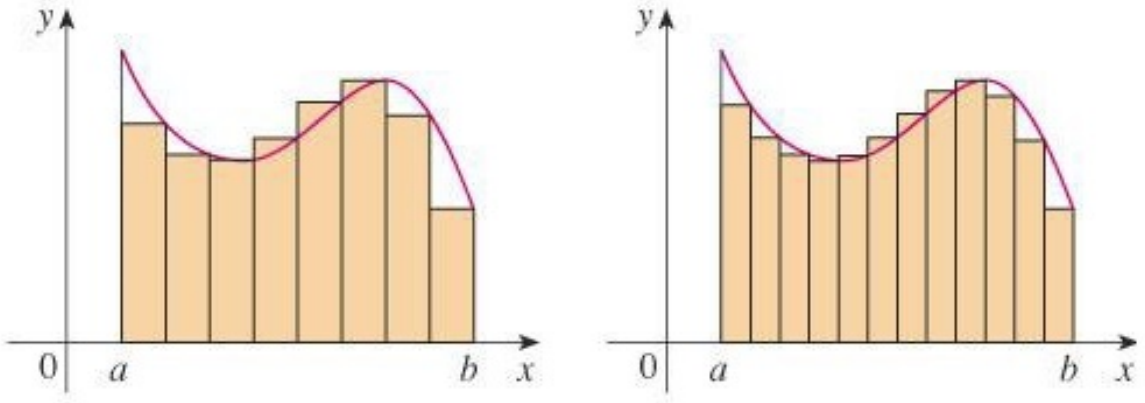
By picking sampling points $x_1^* \in [a, x_1], x_2^* \in [x_1, x_2], \cdots$ in each subinterval, we can approximate the entire area by

$$\sum_{i=1}^{n} f(x_i^*) \cdot \frac{b-a}{n} = \frac{b-a}{n} \cdot \sum_{i=1}^{n} f(x_i^*).$$

Note that as a sampling point, it is most commonly to take

- the left-endpoint $x_i^* = x_{i-1}$, or
- the right-endpoint $x_i^* = x_i$, or
- the midpoint $x_i^* = \dfrac{x_{i-1} + x_i}{2}$.

In general, the larger the n, the smaller the width of each rectangle, and the better the approximation:



(c) $n = 8$ (d) $n = 12$

Therefore, it seems reasonable to define the area under the curve as the limit of the value $\frac{b-a}{n} \cdot \sum_{i=1}^{n} f(x_i^*)$ as $n \to \infty$.

Think of this as a process used to define the value of the area under a curve, rather than a process to compute the area.

Definition. Let $f : [a, b] \to \mathbb{R}$.

- A Riemann sum is the value of $\frac{b-a}{n} \cdot \sum_{i=1}^{n} f(x_i^*)$, where the points $a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$ divide up the interval $[a, b]$ into n equal subintervals, and $x_i^* \in [x_{i-1}, x_i]$ are sampling points.

- The Riemann integral of the function $f(x)$ from a to b is the limit of the Riemann sums as $n \to \infty$. That is

$$\int_a^b f(x)\, dx = \lim_{n\to\infty} \frac{b-a}{n} \cdot \sum_{i=1}^{n} f(x_i^*).$$

If this limit exists and is finite, then we say that $f(x)$ is integrable on the interval $[a, b]$.

Some remarks.

- The value $\int_a^b f(x)\,dx$ is a real number (if it exists) and is defined as the limit of the Riemann sums. It is also called the definite integral of $f(x)$ from a to b.
- The definition of the definite integral works for $a < b$. However if $a = b$ or $a > b$ we adopt the following convention:

$$\int_a^a f(x)\,dx = 0, \qquad \int_a^b f(x)\,dx = -\int_b^a f(x)\,dx.$$

- The variable of integration is not important. So

$$\int_1^5 f(x)\,dx = \int_1^5 f(t)\,dt = \int_1^5 f(z)\,dz.$$

- $\int_a^b f(x)\,dx$ gives the area of the region below the curve $y = f(x)$ from $x = a$ to $x = b$, provided that $f(x) \geq 0$. (**What happens if $f(x) \leq 0$?**)
- **Every integrable function must be bounded on $[a, b]$. There are examples of unbounded and hence non-integrable functions.**

Example

Using the definition of the integral, find

$$\int_0^1 x^2\,dx.$$

Solution. We are asked to use the definition, which says that

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \frac{b-a}{n} \cdot \sum_{i=1}^{n} f(x_i^*).$$

So we have to find the values of a, b, $f(x)$, and x_i^*.

Clearly $a = 0$ and $b = 1$ and $f(x) = x^2$. What about the sample point x_i^*? We can choose x_i^* to be any point in this interval. We pick the right-endpoint, $x_i^* = \frac{i}{n}$.

So plugging in, we get

$$\begin{aligned}
\int_0^1 x^2\,dx &= \lim_{n\to\infty} \frac{1-0}{n} \cdot \sum_{i=1}^{n} \left(\frac{i}{n}\right)^2 \\
&= \lim_{n\to\infty} \frac{1}{n} \cdot \sum_{i=1}^{n} \frac{i^2}{n^2} = \lim_{n\to\infty} \frac{1}{n^3} \cdot \sum_{i=1}^{n} i^2 \\
&= \left(\text{Sum of squares formula } \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}\right) \\
&= \lim_{n\to\infty} \frac{n(n+1)(2n+1)}{6n^3} \\
&= \lim_{n\to\infty} \frac{1\cdot(1+\frac{1}{n})(2+\frac{1}{n})}{6} = \frac{1\cdot(1+0)(2+0)}{6} = \frac{1}{3}.
\end{aligned}$$

• The definition of the definite integral can also sometimes be applied "backwards". That is, you are given the expression involving the Riemann sum and asked to find the exact value of the limit.

Example

Find the limit $\lim_{n\to\infty} \frac{1}{\sqrt{n^3}} \sum_{i=1}^{n} \sqrt{i}$.

Solution. This limit is quite difficult to evaluate. Let's use the definition of the definite integral.

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \frac{b-a}{n} \cdot \sum_{i=1}^{n} f(x_i^*).$$

In the example above, the values of a, b, f are all given in the question.

Now we have to find a suitable combination of a, b, f, x_i^* so that the resulting Riemann sum is equal to the given expression, i.e.

$$\lim_{n\to\infty} \frac{b-a}{n} \cdot \sum_{i=1}^{n} f(x_i^*) = \lim_{n\to\infty} \frac{1}{\sqrt{n^3}} \sum_{i=1}^{n} \sqrt{i}.$$

To do this, we start with the given expression and try to simplify:

$$\begin{aligned} \lim_{n\to\infty} \frac{1}{\sqrt{n^3}} \sum_{i=1}^{n} \sqrt{i} &= \lim_{n\to\infty} \frac{1}{n\sqrt{n}} \sum_{i=1}^{n} \sqrt{i} \\ &= \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} \sqrt{\frac{i}{n}} = \lim_{n\to\infty} \frac{1-0}{n} \sum_{i=1}^{n} \sqrt{\frac{i}{n}}. \end{aligned}$$

Now it is clear that we should choose $a = 0, b = 1$.

What about x_i^*? We pick x_i^* to be the right-endpoint of the interval $\left[\frac{i-1}{n}, \frac{i}{n}\right]$, i.e. $x_i^* = \frac{i}{n}$.

Lastly what about $f(x)$? Clearly we should take $f(x) = \sqrt{x}$ so that

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \frac{b-a}{n} \cdot \sum_{i=1}^{n} f(x_i^*) = \lim_{n\to\infty} \frac{1-0}{n} \sum_{i=1}^{n} \sqrt{\frac{i}{n}}.$$

Now to evaluate this expression, we find $\int_a^b f(x)\,dx = \int_0^1 \sqrt{x}\,dx$ instead. So we conclude that

$$\lim_{n\to\infty} \frac{1}{\sqrt{n^3}} \sum_{i=1}^{n} \sqrt{i} = \int_0^1 \sqrt{x}\,dx.$$

Later we'll see what the value is.

Outline

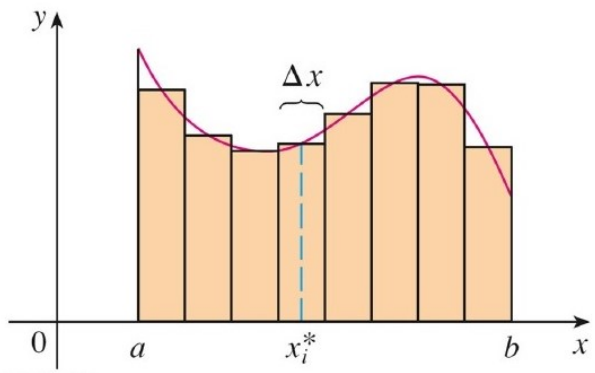
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In this section you study the following.

- How the area is related to the definite integral.
- Basic properties of the definite integral.
- Comparison properties of the definite integral.
- Mean Value Theorem for the definite integral.

How is the area under a graph related to the definite integral?

- If $f \geq 0$ lies above the x-axis from a to b, then the area is simply the definite integral $\int_a^b f(x)\,dx$:

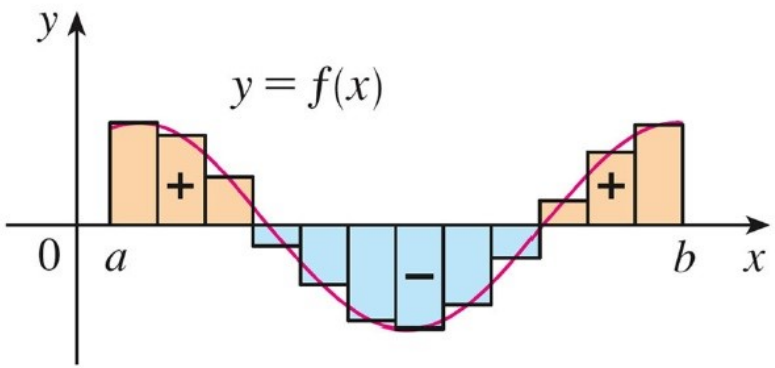


However in the definition of integral, the function f is not restricted to being positive.

- In this case, what value does the definite integral represent?

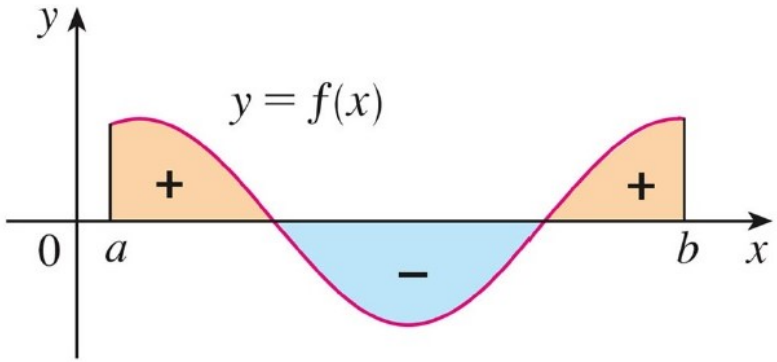
The items of the Riemann sum $\frac{b-a}{n}\sum f(x_i^*)$ are sometimes positive and sometimes negative.

- It is negative if $f(x_i^*) < 0$. As a result, the Riemann sums does not actually represent the sums of areas of the rectangles (see below: the Riemann sums are negative if $f(x_i^*) < 0$).



- So the Riemann sum represents the sum of the areas of the rectangles above the x-axis minus the sum of areas of rectangles below the x-axis.

Let's analyze the figure below about the area of region bounded by the curve $y = f(x)$ and the x-axis



Clearly, after taking limits, the definite integral $\int_a^b f(x)\,dx$ represents the quantity obtained by taking the first and third areas minus the middle area.

So if the question asks you to find the area of region bounded by the curve $y = f(x)$ and the x-axis, we must first find the roots of the function $f(x) = 0$, and then split the integral up accordingly.

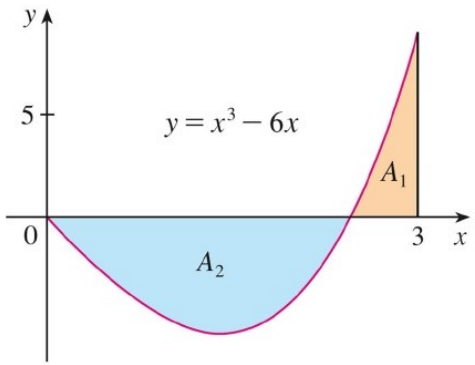
Example

Find the area of the region bounded by the curve $y = x^3 - 6x$ and the x-axis between $x = 0$ to $x = 3$.

Solution. First, we solve for the roots

$$x^3 - 6x = 0 \iff x(x^2 - 6) = 0 \iff x = 0, \pm\sqrt{6}.$$

The region bounded by the curve $y = x^3 - 6x$ and the x-axis between $x = 0$ to $x = 3$ is as follows



- If we simply take the definite integral, we get $\int_0^3 (x^3 - 6x)\, dx = -6.75$, which is negative (is it surprising that we get a negative answer?)

- The correct method is to split the region into A_1 and A_2.
  * The function is negative in the region A_2, and so
    $$A_2 = -\int_0^{\sqrt{6}} (x^3 - 6x)\, dx.$$
  * The function is positive in the region A_1, and so
    $$A_1 = \int_{\sqrt{6}}^3 (x^3 - 6x)\, dx.$$

Thus the final area is $A_1 + A_2$.

Theorem (Basic properties of the definite integral)

Let $f(x)$ and $g(x)$ be integrable function on the interval $[a, b]$, c and d constants. Then

1. $\int_a^a f(x)\,dx = 0$ (Identical limits)
2. $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$ (Reversing limits)
3. $\int_a^b c\,dx = c(b-a)$ (Area of a rectangle)
4. $\int_a^b [cf(x) \pm dg(x)]\,dx = c\int_a^b f(x)\,dx \pm d\int_a^b g(x)\,dx$ (Linearity)
5. $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$
   This holds regardless the order of a, b, c (Additivity)

We also have some further properties of the definite integral.

Theorem (Comparison properties)

Let $f(x)$ and $g(x)$ be integrable on the interval $[a, b]$. TFAT:

1. *If $f(x) \geq 0$ for all $x \in [a, b]$, then $\int_a^b f(x)\,dx \geq 0$.*
2. *If $f(x) \leq g(x)$ for all $x \in [a, b]$, then $\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx$.*
3. *If $m \leq f(x) \leq M$ for all $x \in [a, b]$, then*

$$m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a).$$

This holds even for negative values of m, M.

The following is a very important theorem.

Theorem (Existence of the definite integral)

Every continuous function on a closed interval $[a, b]$ *is integrable.*

So if a function $f(x)$ has finitely many discontinuities, say at the points $a_1, a_2, a_3 \in [a, b]$, we can still compute the integral by taking

$$\int_a^b f(x)\,dx = \int_a^{a_1} f(x)\,dx + \int_{a_1}^{a_2} f(x)\,dx + \int_{a_2}^{a_3} f(x)\,dx + \int_{a_3}^{b} f(x)\,dx.$$

Mean Value Theorem for Integrals

If f is continuous on $[a, b]$, then there exists some point $c \in [a, b]$ such that

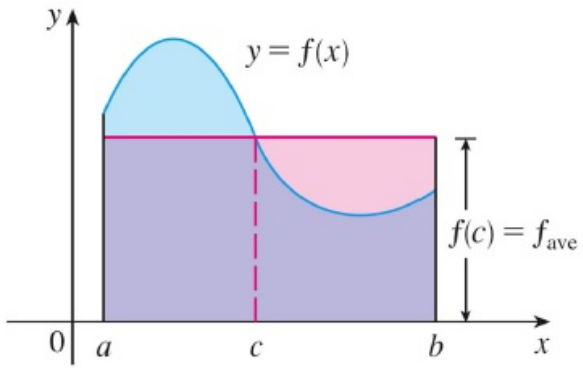
$$f(c) = \frac{1}{b-a} \int_a^b f(x)\, dx.$$

The value $f(c)$ is called the average value of f over the interval $[a, b]$.

Think of what happens if f is the graph of speed versus time. Then $\int_a^b f(x)\, dx$ is the total distance traveled over the time interval $[a, b]$, and $\frac{1}{b-a} \int_a^b f(x)\, dx$ is the average speed over the entire journey.

The MVT says that there is some time $x = c$ where the instantaneous speed $f(c)$ is equal to the average speed over the entire journey.

The average value of f on an interval $[a, b]$



Graphically, the MVT says that we can find some $c \in [a, b]$ such that the rectangle has the same area as the area under the graph. This means that

$$(b - a)f(c) = \int_a^b f(x)\, dx,$$

which is the same as in the theorem.

Example

Show that for any number $t > 0$ there exists a number $0 \leq c \leq t$ such that

$$\cos c = \frac{\sin t}{t}.$$

Solution. Let $f(x) = \cos x$. Then f is continuous and hence integrable on an interval $[0, t]$.

Applying the MVT for Integrals to $f(x)$ and the interval $[0, t]$, we conclude that there is some $c \in [0, t]$, such that

$$\cos c = \frac{1}{t - 0} \int_0^t \cos x \, dx = \frac{\sin t}{t}.$$

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Fundamental Theorem of Calculus - 1

Suppose that f is continuous on $[a, b]$, and define the function

$$F(x) = \int_a^x f(t)\, dt.$$

Then $F'(x) = f(x)$ for any $x \in (a, b)$.

Some remarks.

- For each fixed value of x, the quantity $\int_a^x f(t)\, dt$ is a fixed real number. However if we let x vary then this defines a function of x. So the role of x and the role of t are very different.

- The FTC-1 is usually written in the form

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x).$$

It is important that all three occurrences of the variable x are consistent.

- To apply the FTC-1, the lower limit a must be a constant. If it is not a constant, we must split the integral so that we end up with a constant. In applying the FTC-1, the value of the constant a is irrelevant. So, we have

$$\frac{d}{dx}\int_0^x f(t)\,dt = \frac{d}{dx}\int_{10}^x f(t)\,dt = f(x).$$

Example

Find

(a) $\dfrac{d}{dx}\displaystyle\int_0^{x^2} f(t)\,dt$; (b) $\dfrac{d}{d\sqrt{x}}\displaystyle\int_0^{x} f(t)\,dt$

Solution.

(a) We use the Chain Rule of differentiation. Let $y = x^2$, and we get:

$$\begin{aligned}\frac{d}{dx}\int_0^{x^2} f(t)\,dt &= \left(\frac{d}{dy}\int_0^{y} f(t)\,dt\right)\cdot\frac{dy}{dx}\\ &= f(y)\cdot 2x = 2xf(x^2).\end{aligned}$$

(b) We put $z = \sqrt{x}$ and so we have

$$\begin{aligned}\frac{d}{d\sqrt{x}}\int_0^{x} f(t)\,dt &= \frac{d}{dz}\int_0^{z^2} f(t)\,dt\\ &\underbrace{=}_{\text{by part (a)}} 2zf(z^2) = 2\sqrt{x}f(x).\end{aligned}$$

Now look back at the FTC-1, we can write

$$\left(\int_a^x f(t)\, dt \right)' = f(x),$$

and it shows that if you first integrate, then differentiate the result, you get back $f(x)$.

What happens if you first differentiate, then integrate?

Answer: it is also the same

$$\int_a^x f'(t)\, dt = f(x) + C.$$

FTC - Part 2

Fundamental Theorem of Calculus - 2

If F is an anti-derivative of a continuous function f, then

$$\int_a^b f(x)\,dx = F(b) - F(a) = \left[\int f(x)\,dx\right]_a^b.$$

The FTC-2 is very powerful. It tells us that to compute the value of the definite integral, we do not need to find and use the Riemann sums:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \frac{b-a}{n} \cdot \sum_{i=1}^{n} f(x_i^*).$$

Instead, we only need to find some function F with $F'(x) = f(x)$ and calculate $F(b) - F(a)$.

Example

Find the area of a figure bounded by the line $y = 0$ and the curve $y = (x-2)^2(x+2)^2$.

Solution. The roots of the equation

$$(x-2)^2(x+2)^2 = 0 \iff x = \pm 2,$$

and since $y = (x-2)^2(x+2)^2 \geq 0$, the curve lies above the x-axis between $x = -2$ and $x = 2$.

Thus, the area is simply the definite integral

$$\int_{-2}^{2} (x-2)^2(x+2)^2 \, dx = \int_{-2}^{2} (x^4 - 8x^2 + 16) \, dx.$$

Instead of evaluating the Riemann sums and then taking limits, we use the FTC-2.

Clearly, the anti-derivative of $x^4 - 8x^2 + 16$ is

$$F(x) = \frac{x^5}{5} - \frac{8}{3}x^3 + 16x,$$

which, by the FTC-2, gives

$$\int_{-2}^{2} (x^4 - 8x^2 + 16)\, dx = F(2) - F(-2) = 2\left(\frac{32}{5} - \frac{64}{3} + 32\right).$$

We can also note that the integrand is an even function, and hence

$$\int_{-2}^{2} (x^4 - 8x^2 + 16)\, dx = 2\int_{0}^{2} (x^4 - 8x^2 + 16)\, dx = 2[F(2) - F(0)] = 2F(2).$$